

VOLATILITY MODELS

Nelson R. Ramírez-Rondán

CEMLA

July, 2024

Outline

- 1 What is Volatility?
- 2 Why is Volatility Important?
- 3 Some Volatility Measures
- 4 Implied Volatility
- 5 ARCH Models
- 6 GARCH Models
- 7 Stochastic Volatility Models

What is Volatility?

Volatility is a statistical measure of the dispersion of data around its mean over a certain period of time

In finance, volatility (denoted by σ^2) is the degree of variation of a trading price series over time, usually measured by the standard deviation of logarithmic returns

Estimation Approaches

- Historical volatility measures
- ARCH Models
- GARCH Models
- GARCH Generalizations
- Stochastic Volatility (SV) Models
- Implied Volatility

Why is Volatility Important?

Volatility measurement and estimation are central to finance

- 1 **Asset pricing.** The conditional CAPM involves the conditional second moments of individual asset returns and market returns

$$E_{t-1}(r_{it}) - r_f = \beta_{it} r_{mt}$$

$$\beta_{it} = \frac{\text{cov}_{t-1}(r_{it}, r_{mt})}{\text{var}_{t-1}(r_{mt})}$$

- 2 **Risk management or Value at Risk.** A well known measure of risk called the Value at Risk) can be written as follows

$$\text{VaR}_t(\alpha) = \mu_t + \sigma_t \times q_\alpha$$

where μ_t is the conditional mean and σ_t is the conditional standard deviation of the returns on the asset in question, while q_α is the α -quantile of the innovation distribution

Why is Volatility Important?

- ③ **Portfolio allocation.** The major practical problem that investors have to solve. The most common version of this is the optimization problem

$$\min_{w \in S} w' \text{var}_{t-1}(r_t) w$$

$$\text{s.t. } w' E_{t-1}(r_t) = m$$

where S is some set of portfolio weights that satisfy, $E_{t-1}(r_t)$ is the conditional mean vector and $\text{var}_{t-1}(r_t)$ is the variance matrix

- ④ **Market quality.** volatility can be interpreted as a **measure of the stability** of the underlying system, whereby greater volatility indicates poorer quality

Some Volatility Measures

- It is common to exploit the fact that returns have zero (or close to zero) mean; then, $\sigma^2 \equiv R^2$.
- Historical Average: $\tilde{\sigma}_{t+1}^2 = \frac{1}{t} \sum_{i=1}^t R_i^2$ (full sample)
- Historical Average: $\tilde{\sigma}_{t+1}^2 = \frac{1}{m} \sum_{i=0}^{m-1} R_{t-i}^2$ (moving window of size m)
- Exponential Moving Average: $\tilde{\sigma}_{t+1}^2 = (1 - \lambda)R_t^2 + \lambda\tilde{\sigma}_t^2$; $0 < \lambda < 1$
- Exponential Weighted Moving Average (EWMA):

$$\tilde{\sigma}_{t+1}^2 = \frac{\sum_{i=0}^{m-1} \lambda^i R_{t-i}^2}{\sum_{i=0}^{m-1} \lambda^i}$$

Some Volatility Measures

- Log returns to simplify estimates:

$$R_t = \ln \frac{P_t}{P_{t-1}}$$

$$\sigma = std(R_t) = \sqrt{E[R_t - E(R_t)]^2}$$

$$\hat{\sigma} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2}$$

$vol = \sqrt{252} \hat{\sigma}$ annualized when using daily prices (252 trading days)

$vol = \sqrt{52} \hat{\sigma}$ annualized when using weekly prices (252 trading days)

Implied Volatility from Option Prices

Suppose that stock prices P follow a continuous time model

$$d \ln P(t) = \mu dt + \sigma dB(t)$$

where B is Brownian motion, i.e., for all t, s , $B(t + s) - B(t)$ is normally distributed with mean zero and variance s with independent increments.

The price of many derivatives depend on the underlying asset volatility.

Suppose we have a call option (financial contract that gives the right but not the obligation to buy an asset) on the stock with strike price K (price at which a put or call option can be exercised) and time to maturity τ that we observed. In practice we also observe call option prices C along with the price of the underlying instrument P (an asset on which a derivative contract's price is based).

Implied Volatility from Option Prices

Then we can invert the relation to obtain σ^2 , where we define the value of σ^2 as the value that solves the equation

$$C = C(P, K, r_f, \tau, \sigma^2)$$

In general, this does not yield a closed form solution, but it is a simple problem to compute σ^2 numerically. The quantity σ^2 is called the implied volatility.

One can do this at every time period generating a time series of volatility $\sigma^2(t)$, $t = 1, \dots, T$.

Implied Volatility from Option Prices

If we go back to the Black and Scholes formula, in particular, we saw that the price of the call is given by:

$$C(P(t), t) = N(d_1)P(t) - N(d_2)\exp(-r_f(\tau - t))K$$

$$d_1 = \frac{1}{\sigma(\tau - t)^{1/2}} \left[\ln\left(\frac{S_t}{K}\right) + \left(r_f + \frac{\sigma^2}{2}\right)(\tau - t) \right]$$

$$d_2 = \frac{1}{\sigma(\tau - t)^{1/2}} \left[\ln\left(\frac{S_t}{K}\right) + \left(r_f - \frac{\sigma^2}{2}\right)(\tau - t) \right]$$

where C = call option price. N = CDF of the normal distribution. P_t = spot price of an asset. K = strike price. r_f = risk-free interest rate. τ = time to maturity. σ = volatility of the asset.

Implied Volatility from Option Prices

The VIX index, the so called fear index is the implied volatility of call and puts having as an underlying S&P500 index options with maturities of one month. This is the ticker symbol for the Chicago Board Options Exchange (CBOE) Market Volatility Index.

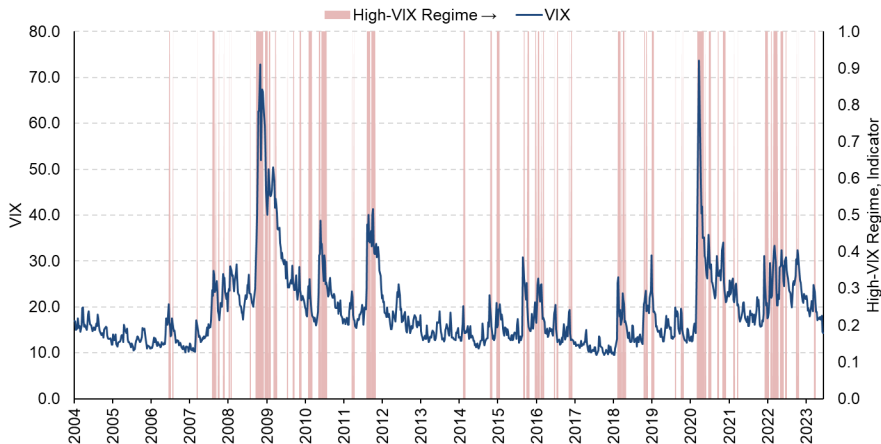
Martin (2017) shows that the population quantity to estimate is

$$VIX_t^2 = \frac{2r_{ft}}{T-t} \left(\int_0^{F_{tT}} \frac{1}{K^2} put_{tT}(K) dK + \int_{F_{tT}}^{\infty} \frac{1}{K^2} call_{tT}(K) dK \right)$$

where T is the time of expiration, F is the forward index level derived from index option prices, and K is the strike price of the put or call option, while put_{tT} and $call_{tT}$ are the put and call option prices.

Implied Volatility from Option Prices

Figure 1: VIX and High-VIX Regime



Note: Monthly averages of VIX index. Red bars indicate weeks with a high volatility regime for the VIX index. The estimation is based on an AR(1) model assuming that the shock's variances are being affected by an underlying Markov regime-switching model. Frequency: Weekly. Last observation: June 14, 2023. Source: With data from Bloomberg.

ARCH Models

Autoregressive conditional heteroscedasticity models (ARCH) are appropriate when the error variance follows an autoregressive (AR) model ([Engle 1982](#)).

An ARCH(p) model has the form:

$$\begin{aligned} R_t &= \mu + \epsilon_t, \text{ mean reverting returns} \\ \epsilon_t &= z_t \sigma_t, \text{ where } z_t \text{ is an iid process} \\ \sigma_t^2 &= \alpha_0 + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2, \text{ with } \alpha_i > 0 \end{aligned}$$

ARCH Models

An ARCH(1) model has the form:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

Next consider the question of stationarity, i.e., whether this process implies that ϵ_t is a stationary process and if so, what the stationary variance is.

Suppose that $\text{var}(\epsilon_t) = \sigma^2 < \infty$. Then it must satisfy

$$\sigma^2 = E(\sigma_t^2) = \alpha_0 + \alpha_1 E(\epsilon_{t-1}^2) = \alpha_0 + \alpha_1 \text{var}(\epsilon_t) = \alpha_0 + \alpha_1 \sigma^2$$

Then, provided $|\alpha_1| < 1$, we get $\sigma^2 = \alpha_0 / (1 - \alpha_1)$.

The equation has this unique solution under the stated condition. On the other hand if $|\alpha_1| > 1$, then σ^2 would be negative, which is not permissible.

GARCH Models

Although the ARCH(1) model implies heavy tails and volatility clustering, it does not in practice generate enough of either. [Bollerslev \(1986\)](#) introduced the generalized autoregressive conditional heteroskedasticity GARCH(p,q) model.

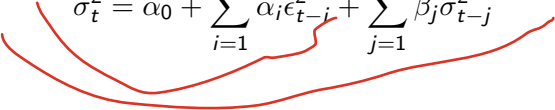
A GARCH(p,q) model, where p is the order of the ARCH terms (lags of ϵ^2) and q is the order of the GARCH terms (lags of σ^2)

Then, a GARCH(p,q) model has the form

ARCH

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

GARCH



GARCH Models

The process ϵ_t is weakly stationary and has finite unconditional variance

$$\sigma^2 = \text{var}(\epsilon_t) = E(\sigma_t^2) = \frac{\alpha_0}{1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j}$$

Provided

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1$$

GARCH Models - Estimation

Under the stationarity condition, σ^2 may be expressed as

$$\sigma^2(\theta) = \frac{\alpha_0}{1 - \sum_{j=1}^q \beta_j} + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \alpha_i \times \sum_{k=1}^{\infty} \sum_{j_1=1}^q \dots \sum_{j_k=1}^q \beta_{j_1} \dots \beta_{j_k} \epsilon_{t-i-j_1-\dots-j_k}^2$$

The Generalized Maximum-Likelihood Estimator (GMLE) is defined as

$$\hat{\theta} = \arg \min_{\theta} \sum_{t=v+1}^T \left[\frac{\epsilon_t^2}{\tilde{\sigma}_t^2(\theta)} + \ln[\tilde{\sigma}_t^2(\theta)] \right]$$

where $\tilde{\sigma}_t^2(\theta)$ is a truncated version of $\sigma^2(\theta)$.

GARCH Models - Estimation

The previous expression depends on the observations $\epsilon_1, \dots, \epsilon_T$ only, and $\nu \geq 1$ is an integer which controls the effect of the truncation.

Note that for a purely ARCH model (i.e. $q = 0$), we choose $\nu = p$.

Note that for $\hat{\epsilon}_t$, we can obtain it by estimating the best fitting of the mean reverting returns $R_t = \mu + \epsilon_t$.

Other GARCH Models

- Integrated Generalized Autoregressive Conditional Heteroskedasticity (IGARCH). The persistent parameters sum up to one. To model level shifts in volatility
- Exponential Generalized Autoregressive Conditional Heteroskedasticity (EGARCH). To allow for asymmetric effects between positive and negative asset returns
- GARCH-in-mean (GARCH-M). The model adds a heteroskedasticity term into the mean equation. To allow that the the return of an asset depends on its volatility (risk)
- Quadratic GARCH (QGARCH). To model asymmetric effects of positive and negative shocks
- Threshold GARCH (TGARCH). To handle leverage effects

Stochastic Volatility Models

The class of stochastic volatility models capture many of the same data features as the GARCH class of models. The model is given by

$$\ln \sigma_t^2 = \alpha_0 + \alpha_1 \ln \sigma_{t-1}^2 + \sigma_\nu \nu_t$$

$$\epsilon_t = z_t \sigma_t \text{ (Recall } R_t = \mu + \sigma_t z_t \text{)}$$

where (z_t, ν_t) are i.i.d. mean zero and variance one and $E(\sigma_t^2) < \infty$.

The main difference from the GARCH class of processes is that there are two shocks z_t, ν_t rather than one.

We distinguish between the case where z_t is independent of ν_t (called the no-leverage case) and the case where this does not hold.

Stochastic Volatility Models

That is ν_t precludes direct observation of the variance σ_t^2

GARCH models are deterministic in the sense that only the mean equation has a disturbance term and that its variance is modeled conditionally on the information up to time $t - 1$. Therefore, the variance can be observed at time t .

$$R_t = \mu + \sigma_t z_t$$

$$R_t = \mu + f(\dots, \nu_t) z_t$$

The overall innovation term of the Stochastic Volatility model depends on ν_t and z_t .

With a zero mean but with a non-Gaussian density.

Estimation methods for Stochastic Volatility Models

In the simplest framework, y_t (or R_t) is modeled as the two stochastic processes.

$$y_t = z_t \sigma_t$$

$$\ln \sigma_t^2 = \phi \ln \sigma_{t-1}^2 + \nu_t$$

where z_t and ν_t are mutually independent Gaussian i.i.d. processes with variances 1 and σ_ν .

Estimation methods: (1) based on y_t and (2) based on $\ln y_t^2$

We distinguish between the case where z_t is independent of ν_t (called the no-leverage case) and the case where this does not hold.

Estimation methods for Stochastic Volatility Models

Methods based on y_t

1. Method of Moments or Generalised Method of Moments

These estimators are based on the convergence of selected sample moments to their unconditionally expected values.

However, these procedures have poor finite sample properties and their efficiency is suboptimal with respect to ML methods ([Jacquier et al., 1994](#)).

optimization fails to converge, specially for small sample sizes ([Andersen and Sørensen, 1996](#)).

They do not generate estimates of the volatility so other procedures as, for example, the Kalman Filter should be used to get those estimates.

Estimation methods for Stochastic Volatility Models

2. Maximum likelihood estimators

ML estimators of the parameters of SV models have experienced a big progress thanks to the development of numerical methods for simulation techniques:

- ① Importance Sampling
- ② Monte Carlo Markov Chain (MCMC)

[Geweke \(1994\)](#) introduces the importance sampling algorithm applied to SV models.

Estimation methods for Stochastic Volatility Models

Methods based on $\log(y_t^2)$

1. Quasi-maximum likelihood (QML)

A quasi-maximum likelihood (QML) a function that is related to the logarithm of the likelihood function, where we assume some parts of the distribution may be mis-specified.

The function that is maximized to form a QMLE is often a simplified form of the actual log likelihood function (e.g. treats certain data values as being independent, even when in actuality they may not be).

The QML estimator is based on linearizing the stochastic volatility model by taking the logarithms of the squares of the observations.

Estimation methods for Stochastic Volatility Models

The function can be written as

$$\log L(\log(y_t^2)|\theta) = -\frac{1}{2} \sum_{t=1}^T \log F_t - \frac{1}{2} \sum_{t=1}^T \frac{\nu_t^2}{F_t},$$

where ν_t is the one-step-ahead prediction error of $\log(y_t^2)$ and F_t is the corresponding mean squared error.

QML estimator is consistent and asymptotically normal. However, the QML procedure is inefficient; but very flexible.

Estimation methods for Stochastic Volatility Models

2. Other methods based in linearization

Simulated expectation maximization (SEM) algorithm.

Gibbs sampling?

References

- Bollerslev, T. (1986). Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics*, 31(3), 307-327.
- Engle, R.F. (1982). Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica* 50(4), 987-1007.
- Linton, O. (2019). *Financial Econometrics Models and Methods*. Cambridge University Press.
- Martin, I. (2017). What is the Expected Return on the Market? *Quarterly Journal of Economics*, 132(1), 367-433.